University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

First Midterm Exam – Version A

October 11, 2012

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	4	4	3	3	5	5	24
Grade							

1. [**4 points**] Determine if the following linear system is consistent or inconsistent. (If the system is consistent, you do not need to find the general solution.)

$$\begin{cases} x_1 + 2x_2 = x_3 - 5x_4 + 1 \\ \quad -2x_2 = 4x_4 + 1 \\ 2x_1 + 2x_2 = 2x_3 - 6x_4 + 4 \end{cases}$$

2. [**4 points**] Determine all values of h such that the vector $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ belongs to Span $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ h \\ 1 \end{bmatrix} \right\}$.

3. [**3 points**] Find the general solution for the matrix equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ -2 & 4 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}.$$

You should write your solution in **vector parametric form**.

4. [**3 points**] For which values of the parameters a and b the linear system given below is inconsistent?

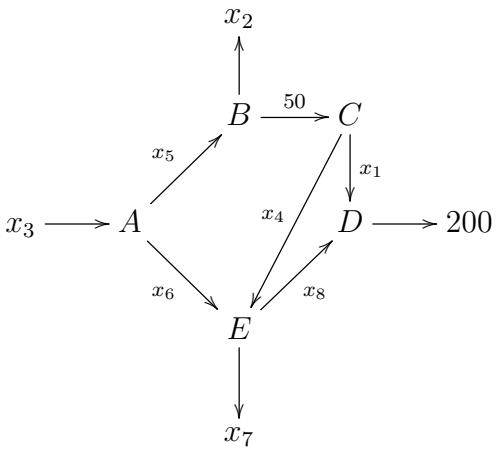
$$\begin{cases} x_1 + 2x_2 = 1 \\ x_2 - 2x_3 = 0 \\ -2x_1 + ax_2 - x_3 = b \end{cases}$$

5. [**5 points**] Determine if the following vectors are linearly dependent or linearly independent:

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix}$$

If the vectors are linearly dependent, then you should find a (nontrivial) linear combination of them which is equal to the zero vector.

6. Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) **[3 points]** Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 8$. (Do not solve the linear system at this stage.)

(b) [**1 point**] The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{cccccccc|c} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 200 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & 0 & -1 & 0 & -1 & 0 & 150 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & -150 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 150 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

(c) [**1 point**] Suppose that due to road work, the flow along ED is limited to a maximum of 200 cars per minute. What is the maximum possible flow along CE?

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